705.641.81: Natural Language Processing Self-Supervised Models

Homework 5: RNNs and Transformers/Self-Attention

For homework deadline and collaboration policy, please see our Canvas page.

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Collaborators, if any: Yuzi Li

Sources used for your homework, if any: _____

This assignment focuses on basic understanding of Recurrent Neural Networks and Transformers for language modeling. We will also touch upon Tokenization.

Homework goals: After completing this homework, you should be comfortable with:

- aligning your prior understanding of Neural Net training for RNNs;
- thinking about Self-Attention module and implementing a simple Transformer language model.

1 Concepts, intuitions and big picture

Each question might have multiple correct answers. (Check all that apply).

1. Which of these types of models would you use for completing prompts with generated text?
   ☐ An encoder model
   ☒ A decoder model

2. Which of these types of models would you use for classifying text inputs according to certain labels?
   ☒ An encoder model
   ☐ A decoder model

3. To which of these tasks would you apply a many-to-one RNN architecture?
   ☐ Speech recognition (input an audio clip and output a transcript)
   ☒ Sentiment classification (input a piece of text and output a 0/1 to denote positive or negative sentiment)
   ☒ Gender recognition from speech (input an audio clip and output a label indicating the speaker's gender)

4. What possible source can the bias observed in a model have?
   ☒ The model is a fine-tuned version of a pretrained model and it picked up its bias from it.
   ☒ The data the model was trained on is biased.
   ☒ The metric the model was optimizing for is biased.

5. How many dimensions does the tensor output by a decoder Transformer model have, and what are they?
   ☐ 2: The sequence length and the batch size
   ☐ 2: The sequence length and the hidden size
   ☒ 3: The sequence length, the batch size, and the hidden size

6. You are training an RNN language model. At the tth time step, what is the RNN doing? Choose the best answer.
   ☐ Estimating $P(y_1, y_2, ..., y_{t-1})$
   ☐ Estimating $P(y_1)$
   ☒ Estimating $P(y_t|y_1, y_2, ..., y_{t-1})$
   ☐ Estimating $P(y_t|y_1, y_2, ..., y_t)$

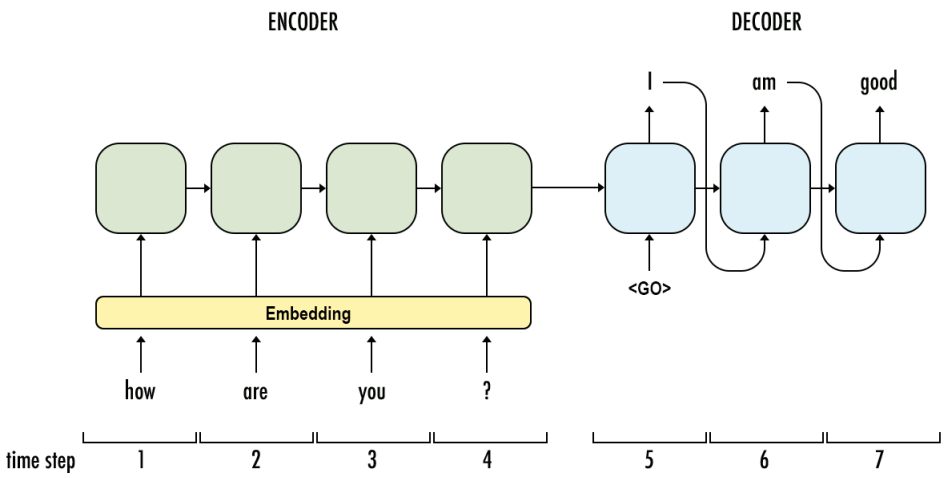
7. You are training an RNN, and find that your weights and activations are all taking on the value of `NaN` ("Not a Number"). Which of these is the most likely cause of this problem?
   ☐ Vanishing gradient problem.
   ☒ Exploding gradient problem.
   ☐ ReLU activation function g(.) used to compute g(z), where z is too large.
   ☐ Sigmoid activation function g(.) used to compute g(z), where z is too large.

8. You're done training an RNN language model. You're usig it to sample random sentences as follows. What are you doing at each time step t?
   ☐ (i) Use the probabilities output by the RNN to pick the highest probability word for that time-step as y_t. (ii) Then pass the ground-truth word from the training set to the next time-step.
   ☐ (i) Use the probabilities output by the RNN to randomly sample a chosen word for that time-step as y_t. (ii) Then pass the ground-truth word from the training set to the next time-step.
   ☐ (i) Use the probabilities output by the RNN to pick the highest probability word for that time-step as y_t. (ii) Then pass this selected word to the next time-step.
   ☒ (i) Use the probabilities output by the RNN to randomly sample a chosen word for that time-step as y_t. (ii) Then pass this selected word to the next time-step.

9. You have a pet crab whose mood is heavily dependent on the current and past few days' weather. You've collected data for the past 42 days on the weather, which you represent as a sequence as $x_1, ..., x_{42}$. You've

also collected data on your crab's mood, which you represent as $y_1, ..., y_{42}$. You'd like to build a model to map from $x \to y$. Should you use a Unidirectional RNN or Bidirectional RNN for this problem?
☐ Bidirectional RNN, because this allows the prediction of mood on day t to consider more information.
☐ Bidirectional RNN, because this allows backpropagation to compute more accurate gradients.
☒ Unidirectional RNN, because the value of y_t depends only on $x_1, ..., x_t$ but not on $x_{t+1}, ..., x_{42}$
☐ Unidirectional RNN, because the value of y_t depends only on x, and not other days' weather.

10. Consider this encoder-decoder model. This model is a "conditional language model" in the sense that the



encoder portion (shown in green) is modeling the probability of the input sentence x.
☐ True ☒ False

11. Compared to the RNN-LMs and fixed-window-LMs, we expect the attention-based LMs to have the greatest advantage when:
☒ The input sequence length is large.
☐ The input sequence length is small.

12. What is a model head?
☐ A component of the base Transformer network that redirects tensors to their correct layers
☐ Also known as the self-attention mechanism, it adapts the representation of a token according to the other tokens of the sequence
☒ An additional component, usually made up of one or a few layers, to convert the transformer predictions to a task-specific output

13. What are the techniques to be aware of when batching sequences of different lengths together?
☒ Truncating ☐ Returning tensors ☒ Padding ☒ Attention masking

2. (Extra Credit) Inferring distr from semi-closed API

$$p(v; b_v) = \frac{e^{f(v)+b_v}}{\tilde{z}} = \frac{e^{f(v)+b_v}}{z + e^{f(v)+b_v} - e^{f(v)}}$$

$$p(v^*; b_v) = \frac{e^{f(v^*)+b_{v^*}}}{\tilde{z}} = \frac{e^{f(v^*)+b_{v^*}}}{z + e^{f(v^*)+b_{v^*}} - e^{f(v^*)}}$$

$$p(v^*) = \frac{e^{f(v^*)+b_{v^*}}}{z}$$

Goal: $p(v) = \dfrac{e^{f(v)}}{z}$

Define $r = \dfrac{p(v^*; b_v)}{p(v^*)} = \dfrac{z}{\tilde{z}}$

$$p(v; b_v) + r = \frac{z + e^{f(v)+b_v}}{\tilde{z}}$$

$$= \frac{\tilde{z} + e^{f(v)}}{\tilde{z}}$$

$$= 1 + \frac{e^{f(v)}}{\tilde{z}}$$

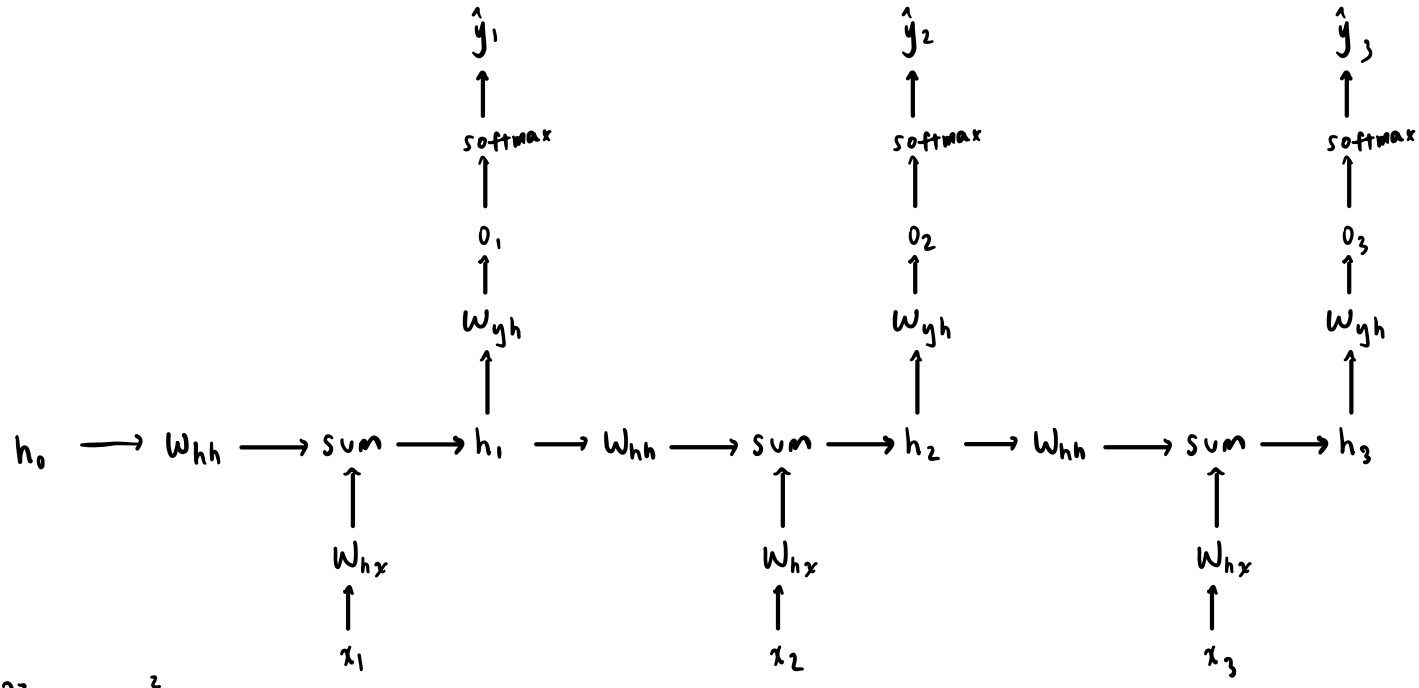
$$\left[p(v; b_v) + r - 1\right] r^{-1} = \left[\frac{e^{f(v)}}{\tilde{z}}\right] \frac{\tilde{z}}{z} = \frac{e^{f(v)}}{z} = P(v)$$

Simplify: $$P(v) = \frac{e^{f(v)}}{z} = \frac{\left[p(v; b_v) + \dfrac{p(v^*; b_v)}{p(v^*)} - 1\right] p(v^*)}{p(v^*; b_v)}$$

$$= \frac{p(v^*)\left(p(v; b_v) - 1\right)}{p(v^*; b_v)} + 1$$

3. Training an RNN

3.1



3.2.1 $$\frac{\partial J_t}{\partial o_{t,j}} = \sum_{k=1}^{2} \frac{\partial J_t}{\partial \hat{y}_{t,k}} \frac{\partial \hat{y}_{t,k}}{\partial o_{t,j}}$$

$$\frac{\partial J_t}{\partial \hat{y}_{t,k}} = \frac{\partial}{\partial \hat{y}_{t,k}} \left(- \sum_{i=1}^{2} y_{t,i} \log(\hat{y}_{t,i}) \right) = - \frac{y_{t,k}}{\hat{y}_{t,k}}$$

$$\frac{\partial \hat{y}_{t,k}}{\partial o_{t,j}} = \begin{cases} \hat{y}_{t,k}(1-\hat{y}_{t,k}) & \text{if } k=j \\ -\hat{y}_{t,k}\,\hat{y}_{tj} & \text{otw} \end{cases}$$

$$\Rightarrow \frac{\partial J_t}{\partial o_{tj}} = -y_{t,j}(1-\hat{y}_{t,j}) + \sum_{k \neq j} y_{t,k}\,\hat{y}_{tj}$$

$$= -y_{tj} + \hat{y}_{tj} \Big(y_{tj} + \underbrace{\sum_{k \neq j} y_{t,k}}_{=1} \Big)$$

$$= \hat{y}_{tj} - y_{tj}$$

$$\rightarrow \frac{\partial J_t}{\partial o_t} = \hat{y}_t - y_t$$

3.2.2 $$\frac{\partial J_t}{\partial h_t} = g_{o_t} \frac{\partial o_t}{\partial h_t} = g_{o_t} W_{yh}$$

$$h_t = \psi(z_t) = W_{hh} h_{t-1} + W_{hx} x_t$$

$$\frac{\partial h_t}{\partial h_{t-1}} = W_{hh} \qquad \frac{\partial h_{t-1}}{\partial h_{t-2}} = W_{hh} \quad \cdots$$

$$\frac{\partial J}{\partial h_i} = g_{o_t} \frac{\partial o_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \cdots \frac{\partial h_{i+1}}{\partial h_i}$$

$$= g_{o_t} W_{yh} (W_{hh})^{t-i}$$

3.2.3 $$\frac{\partial J_t}{\partial W_{hh}} = \sum_{i=1}^{t} g_{h_i} \frac{\partial h_i}{\partial W_{hh}}$$

$$\frac{\partial h_i}{\partial W_{hh}} = h_{i-1}$$

$$\therefore \frac{\partial J_t}{\partial W_{hh}} = \sum_{i=1}^{t} g_{h_i} h_{i-1}^{T}$$

3.2.4 $$\frac{\partial J}{\partial W_{hh}} = \frac{\partial}{\partial W_{hh}} \sum_{t=1}^{3} J_t = \sum_{t=1}^{3} g_{W_{hh},t}$$

4. Self-Attention & Transformers

4.1 $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{h}}\right) V$

Need to compute:

$Q = XW_q$, $K = XW_k$, $V = XW_v$ are each $O(nh^2)$

QK^T is $(n \times h) \times (h \times n)$ is $O(n^2 h)$

$\text{softmax}(\ ) \cdot V$ is $(n \times n) \times (n \times h)$ is $O(n^2 h)$

At test time, h is fixed so complexity grows with n^2; $O(n^2)$.

Since the attention mechanism does not require sequential processing, we can process the input data in parallel with GPUs.

4.2 We can add a mask to the attention scores:

$$A^M = \text{Attention}\left(\frac{QK^T}{\sqrt{h}} + M\right) V$$

M is a matrix that has $-\infty$ for elements of column 3; 0 otw.

(Extra credit)

4.3 SVD gives $P_{n \times n} = U_{n \times k} \Sigma_{k \times k} S^T_{k \times n}$

Let $A = (U\Sigma)_{n \times k}$, $B = (S^T V)_{k \times h}$

We can compute B in $O(nkh)$

$\text{Attention} = PV = \underbrace{AB}_{O(nkh)}$ is $(n \times k) \times (k \times h)$

$\therefore$ runs in $O(nkh) + O(nkh) = O(nkh)$

4.4 $P = \frac{QK^T}{\sqrt{h}} = U_{n \times k} \Sigma_{k \times k} S^T_{k \times n}$

Define $D = S^T_{k \times n}$ is a "projection" onto the principal subspace. Note that $SS^T = I$

(strictly rotation...) and not idempotent

$S^T S$ is a projection matrix

Define $C = \frac{1}{\sqrt{h}} S^T$.

Then $P = PSS^T = \sqrt{h}\, PC^T D$

$\therefore PV = \sqrt{h}\, PC^T D V = \cancel{\sqrt{h}} \frac{QK^T}{\cancel{\sqrt{h}}} C^T D V = QK^T C^T D V = Q(CK)^T \underbrace{DV}_{O(nkh \text{ like before in 4.3})}$

$B := (CK)^T_{h \times k}$ is $((k \times n) \times (n \times h))^T$ is $O(nkh)$

QB is $(n \times h) \times (h \times k)$ is $O(nkh)$

$\therefore O(nkh) + O(nkh) + O(nkh) = O(nkh)$

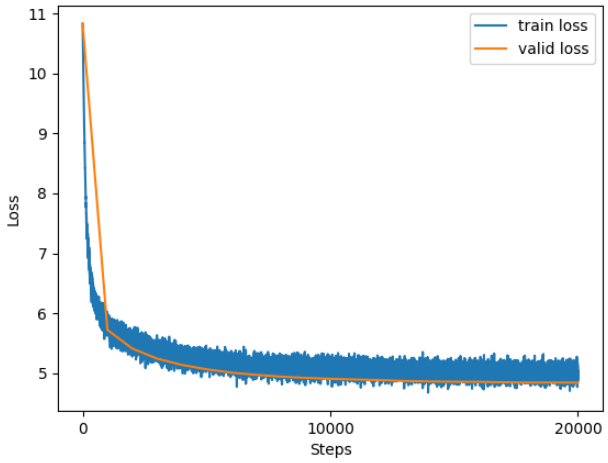
5. Programming

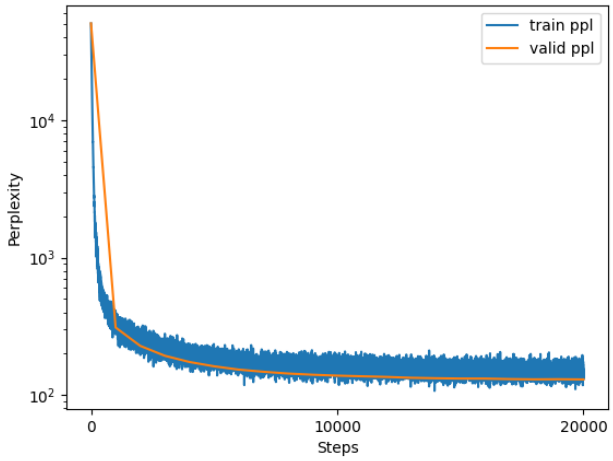
5.1.a

(i) CPU Run time: 408.77s

GPU Run time: 3.84s

(ii)





The best perks of living on the east – TNA , but the team , and led a number of men 's family to move the city 's land in the town .

It is a truth universally acknowledged of his family . The poem , to be an early woman who tells that they had " . According to a woman as well as she tells a relationship

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The best way to learn they are able to have an average day .
are no longer possible , a result of the two @-@ minute , but the team should

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According to the latest research of the University of New Zealand , the National Register of Historic Places in 2006 .
was not included by all the official website . In 2004 2011